Data and Artificial Intelligence
Cyber Shujaa Program

Week 7 Assignment
Regression Models

Student Name: Salome Kungu

Student ID: CS-DA03-26054

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Introduction

The objective of this assignment was to develop hands-on experience in building and evaluating **two regression models using Python and Scikit-learn**:

1. **Simple Linear Regression** (Single independent variable)
2. **Multiple Linear Regression (Multivariate Regression)** (Multiple independent variables)

The goal was to predict a target outcome (such as exam scores or house prices) using one feature in the first model and multiple features in the second model.

The key objectives of the assignment were to:

1. Dataset exploration
2. Data preprocessing and splitting
3. Model training
4. Model evaluation using statistical metrics
5. Visualization of predictions
6. Publishing the project to a professional portfolio

This project strengthened my understanding of regression modeling, feature relationships, model evaluation, and comparative analysis between single-feature and multi-feature models.

Tasks Completed

Link to Code:

https://colab.research.google.com/drive/1F-XVzmsZmk1fDbPMjtAZ035HUpbSIXfb?usp=sharing

LINEAR REGRESSION

Task 1: Load and Explore the Dataset

The dataset was successfully loaded and analyzed. Initial visualizations showed relationships between independent variables and the target variable, supporting the use of regression modeling.

What:

- Select one independent variable
- Plot scatter plot (X vs y)
- Check correlation coefficient

Why:

- To verify linear relationship
- To ensure simple regression is appropriate
- To visually confirm trend direction (positive/negative)

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LINEAR REGRESSION

1. Explore the Dataset

- Loading the data

```
import pandas as pd
import numpy as np
import csv
import warnings
warnings.filterwarnings("ignore")
import matplotlib.pyplot as plt
%matplotlib inline
```

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1. Explore the Dataset

- Loading the data

```
import pandas as pd
import numpy as np
import csv
import warnings
warnings.filterwarnings("ignore")
import matplotlib.pyplot as plt
%matplotlib inline
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
regression = pd.read_csv("/content/drive/MyDrive/Regression Datasets/wk07-regression/homeprices.csv")
```

Salome Kungu_CS-DA03-26054_WK7.ipynb

```
mounted at /content/drive
```

```
regression = pd.read_csv("/content/drive/MyDrive/Regression Datasets/wk07-regression/homeprices.csv")
regression.head()
```

	area	price
0	2600	550000
1	3000	565000
2	3200	610000
3	3600	680000
4	4000	725000

- Data Structure and Summary Statistics

```
print("Regression DataFrame Information:")
```

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- Data Structure and Summary Statistics

```
print("Regression DataFrame Information:")
regression.info()
```

```
Regression DataFrame Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 2 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   area    5 non-null      int64
 1   price   5 non-null      int64
dtypes: int64(2)
memory usage: 212.0 bytes
```

```
print("\nRegression DataFrame Descriptive Statistics:")
regression.describe()
```

Salome Kungu_CS-DA03-26054_WK7.ipynb

```
print("\nRegression DataFrame Descriptive Statistics:")
regression.describe()
```

```
Regression DataFrame Descriptive Statistics:
```

	area	price
count	5.000000	5.000000
mean	3280.000000	626000.000000
std	540.370243	74949.983322
min	2600.000000	550000.000000
25%	3000.000000	565000.000000
50%	3200.000000	610000.000000
75%	3600.000000	680000.000000
max	4000.000000	725000.000000

- Visualization using scatter plots

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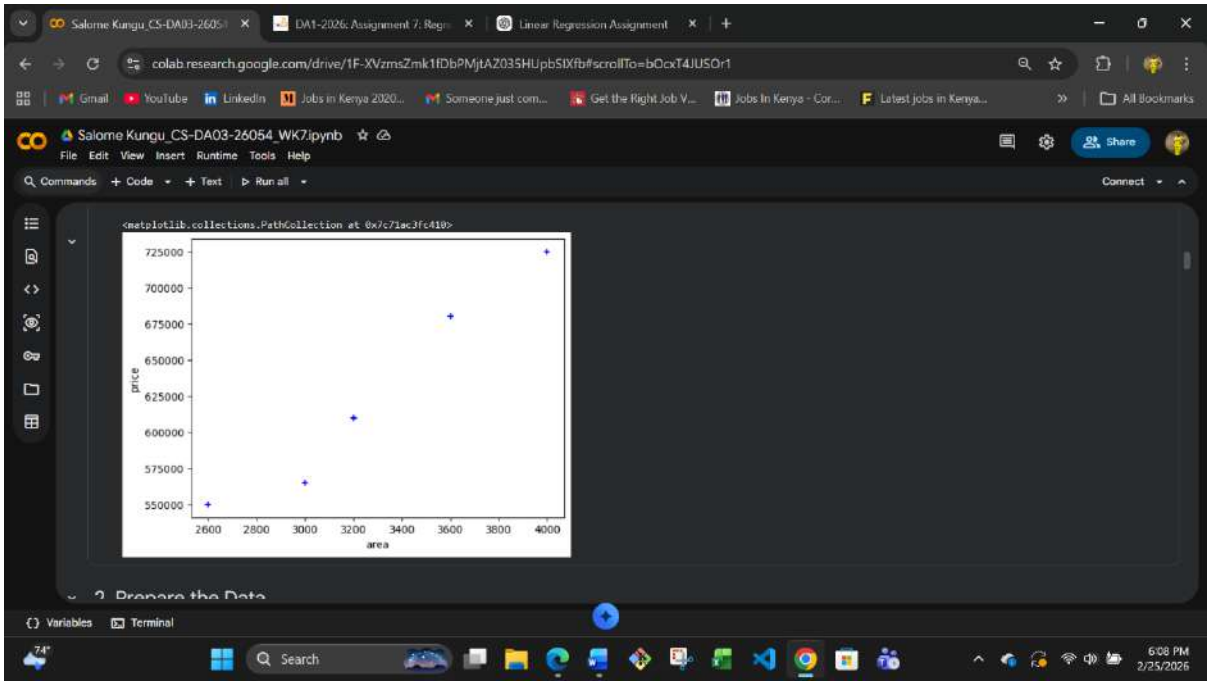
- Visualization using scatter plots

```
%matplotlib inline
plt.xlabel('area')
plt.ylabel('price')
plt.scatter(regression.area,regression.price,color='blue',marker='+')
```

```
<matplotlib.collections.PathCollection at 0x7c71ac3fc410>
```



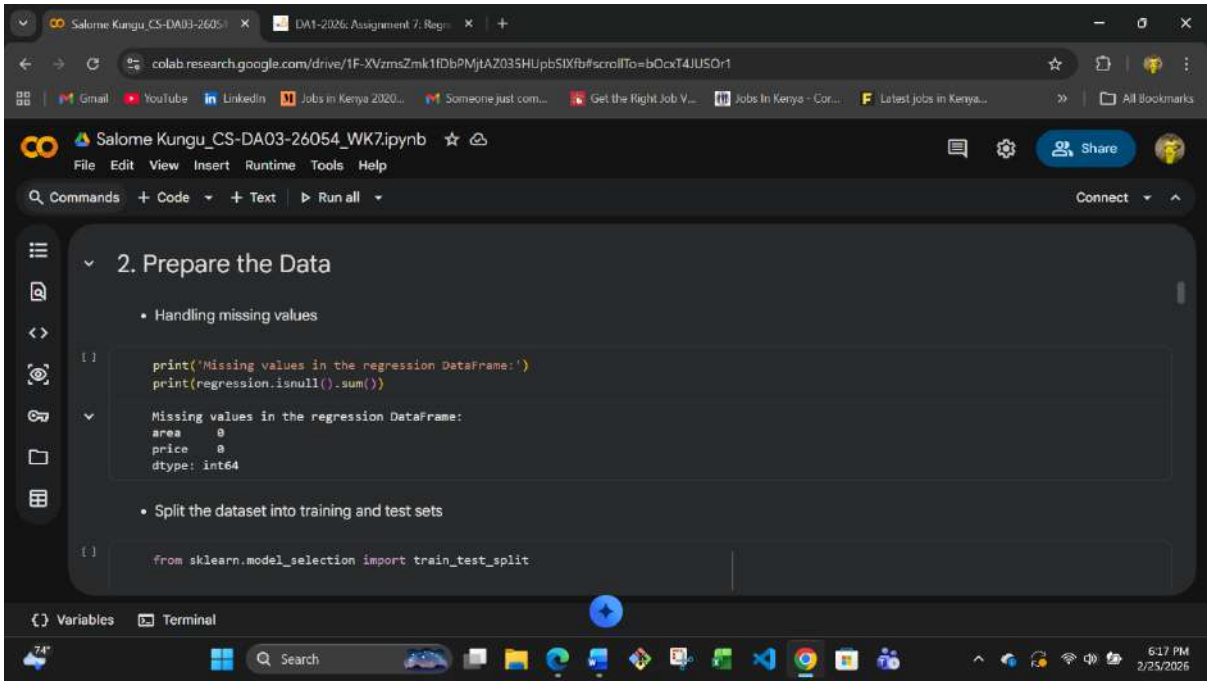
Task 2: Data Preparation for Simple Regression

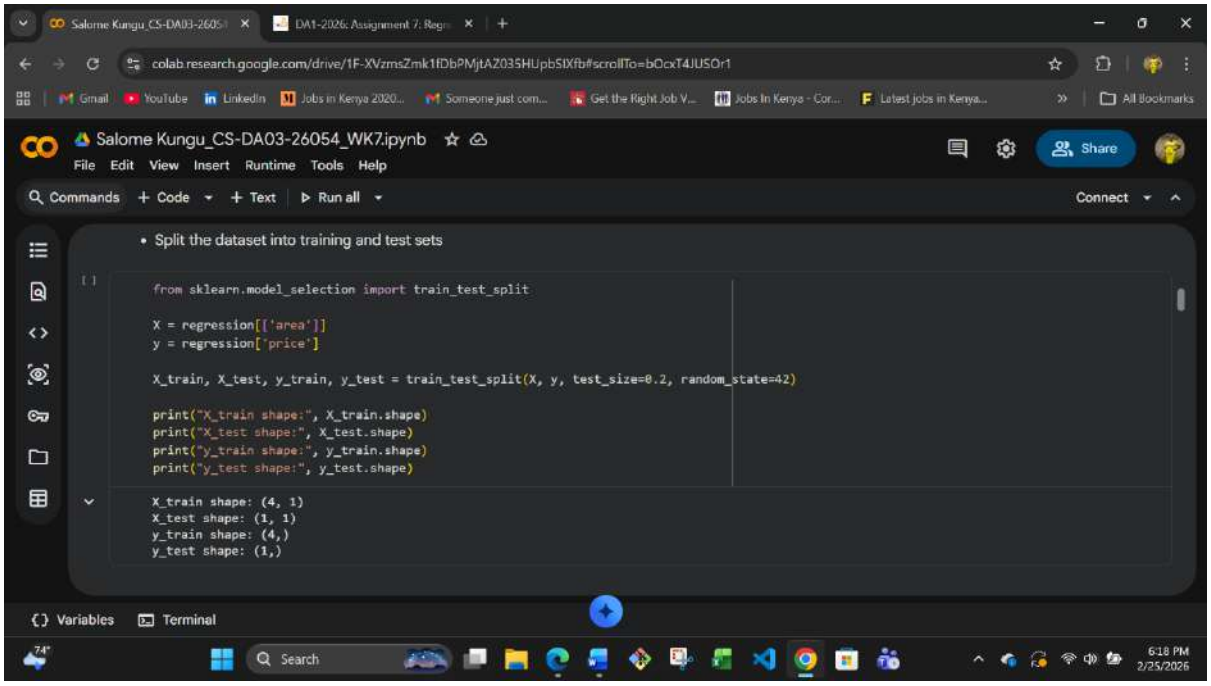
What:

- Perform train-test split

Why:

- To isolate one predictor
- To prepare structured input for regression model
- To prevent overfitting





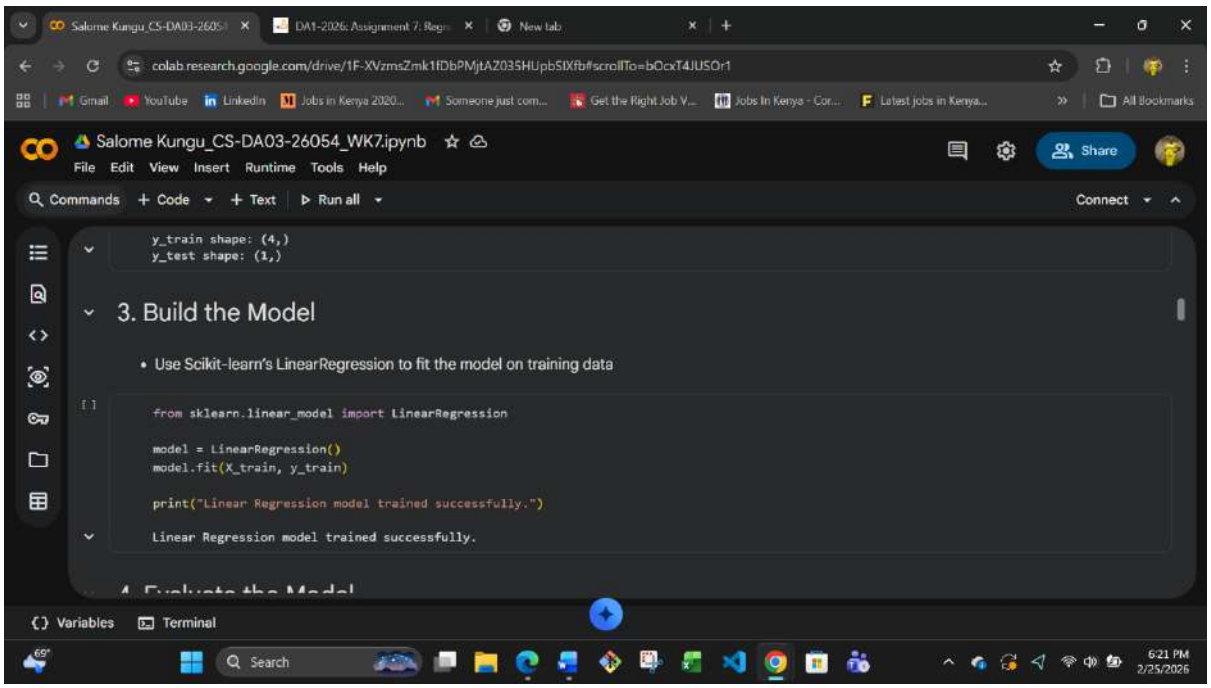
Task 3: Train Simple Linear Regression Model

What:

- Fit LinearRegression() on training data
- Extract coefficient and intercept

Why:

- To compute best-fit straight line
- To quantify impact of single feature
- To build baseline predictive model



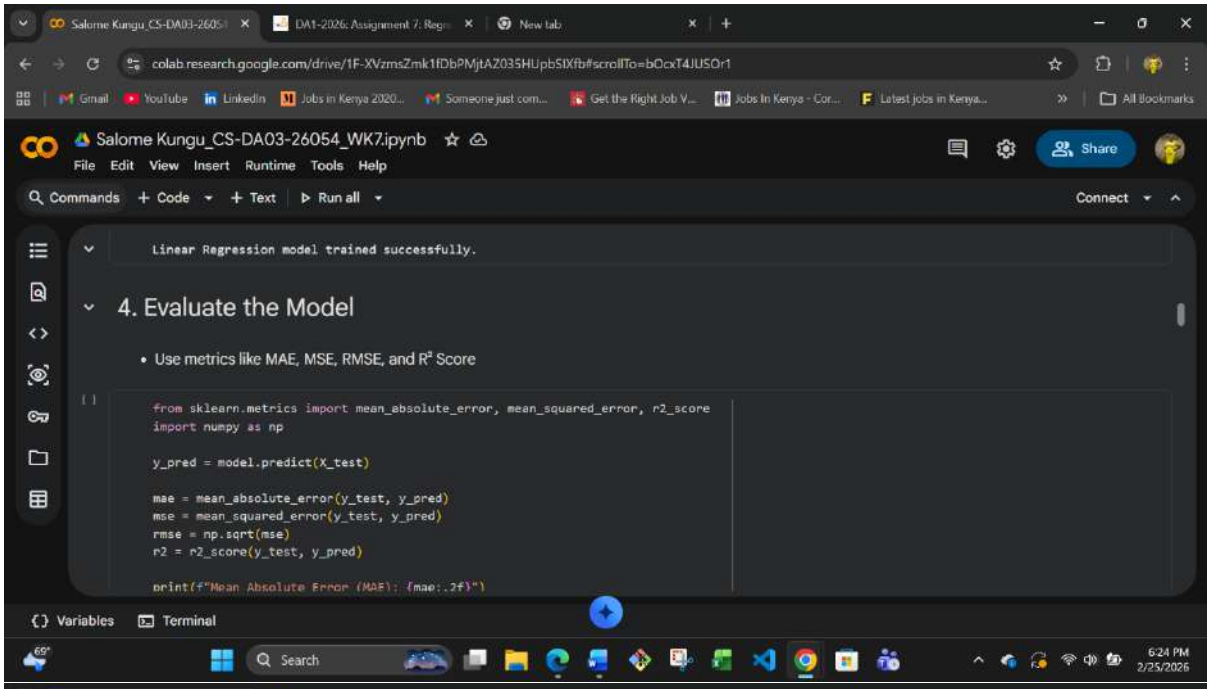
Task 4: Evaluate Simple Linear Regression

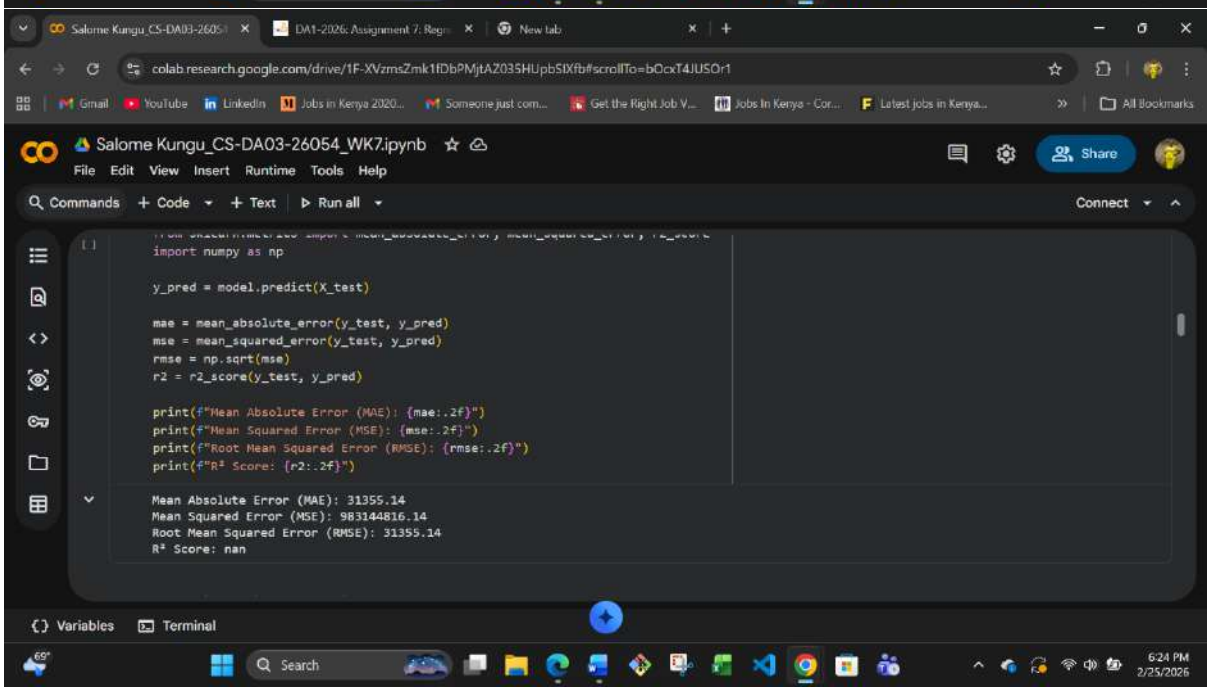
What:

- Compute MAE
- Compute MSE
- Compute RMSE
- Compute R^2

Why:

- To measure prediction accuracy
- To evaluate explanatory power
- To establish performance benchmark





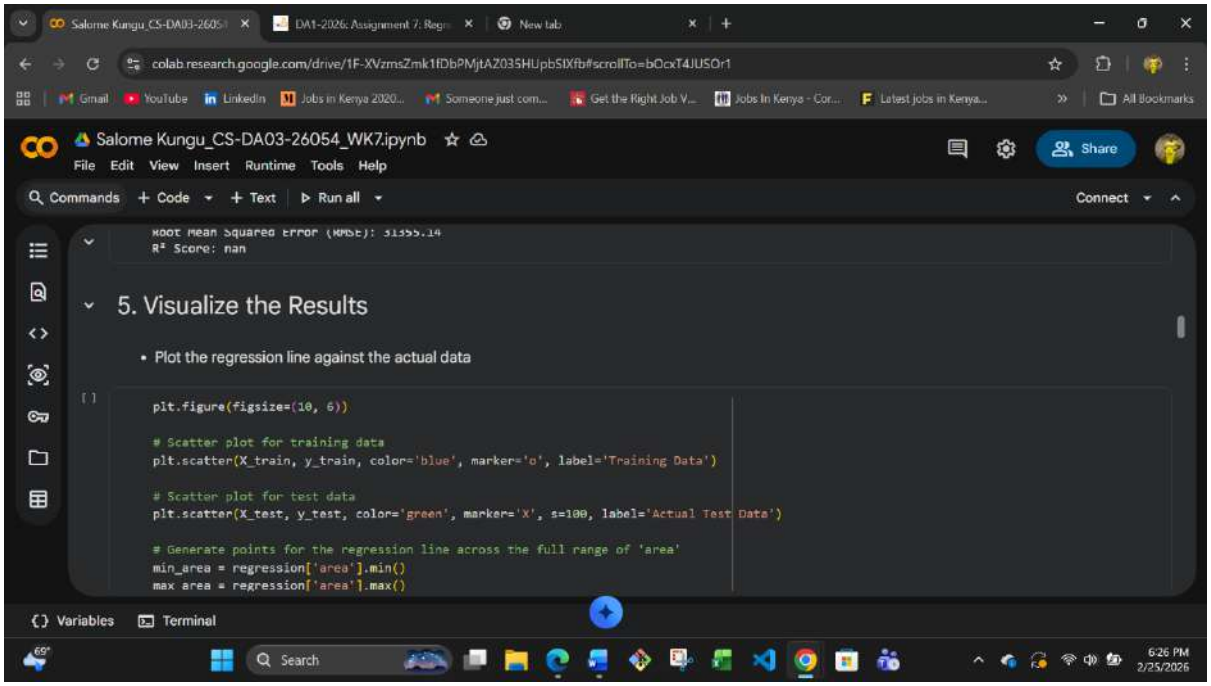
Task 5: Visualize Regression Line

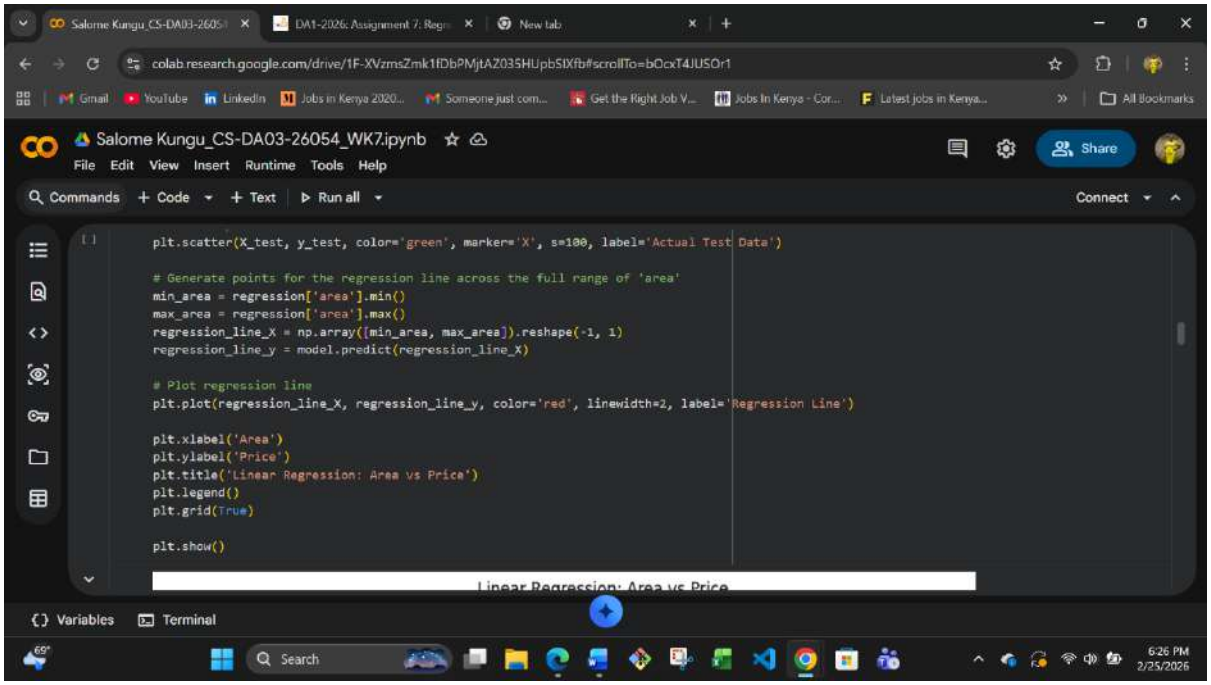
What:

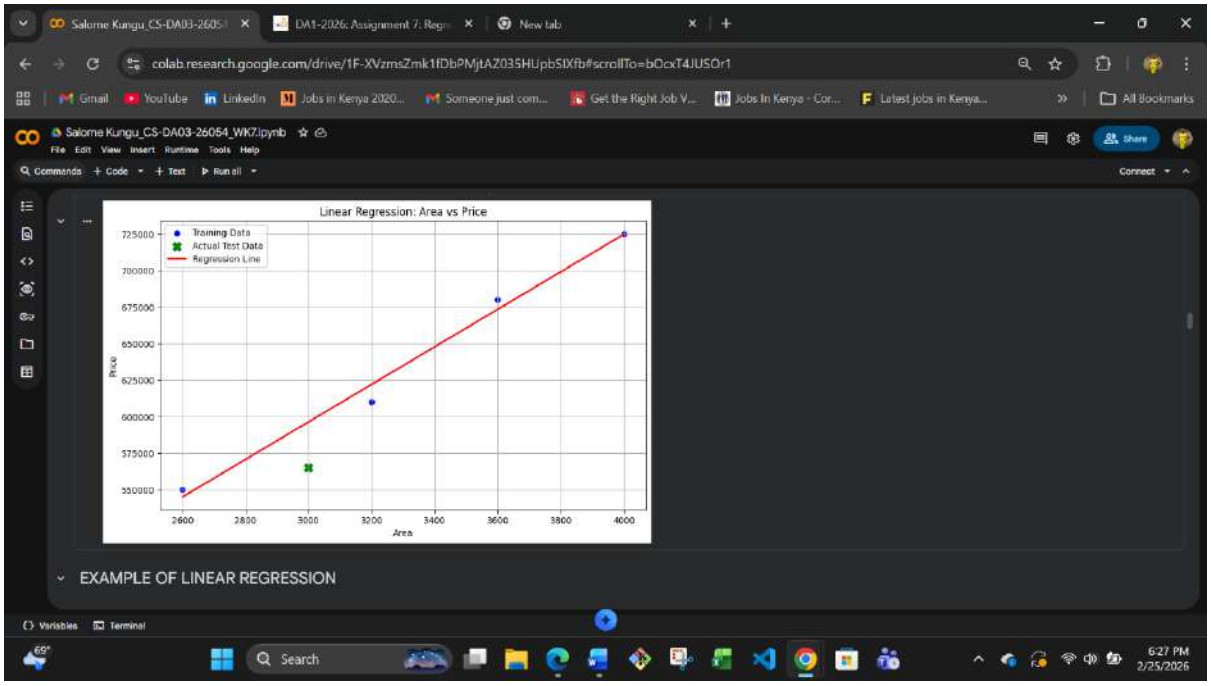
- Plot scatter plot
- Overlay regression line

Why:

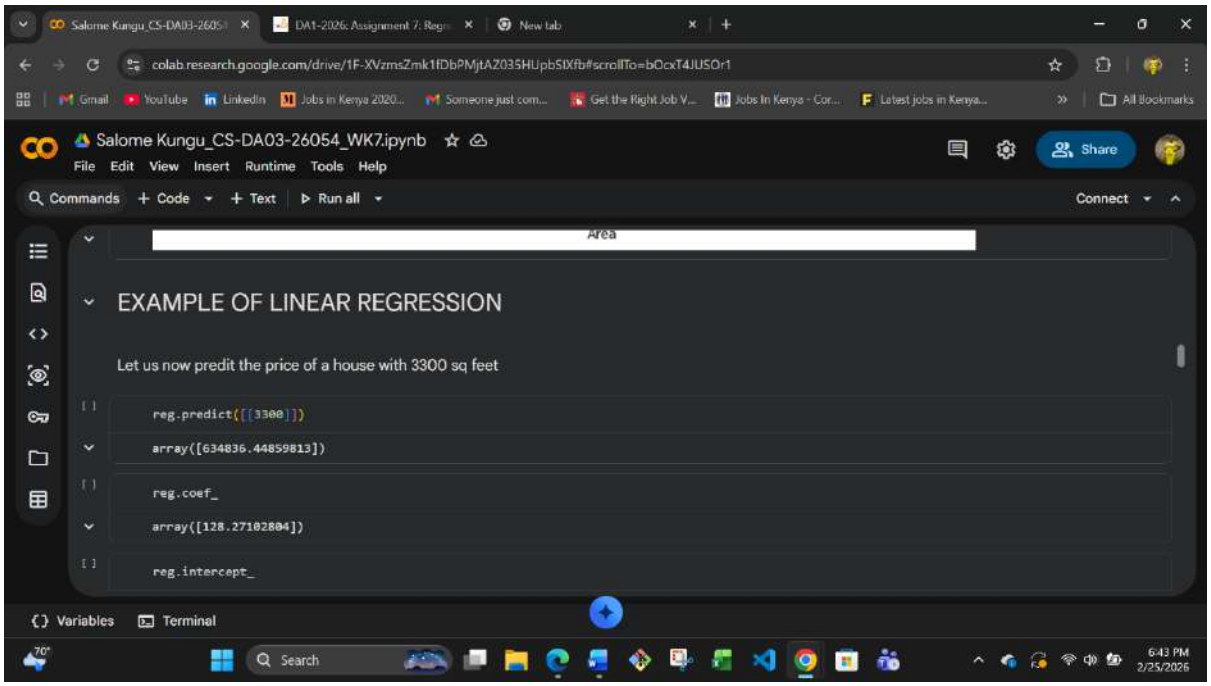
- To visually assess goodness-of-fit
- To detect underfitting
- To interpret model behavior

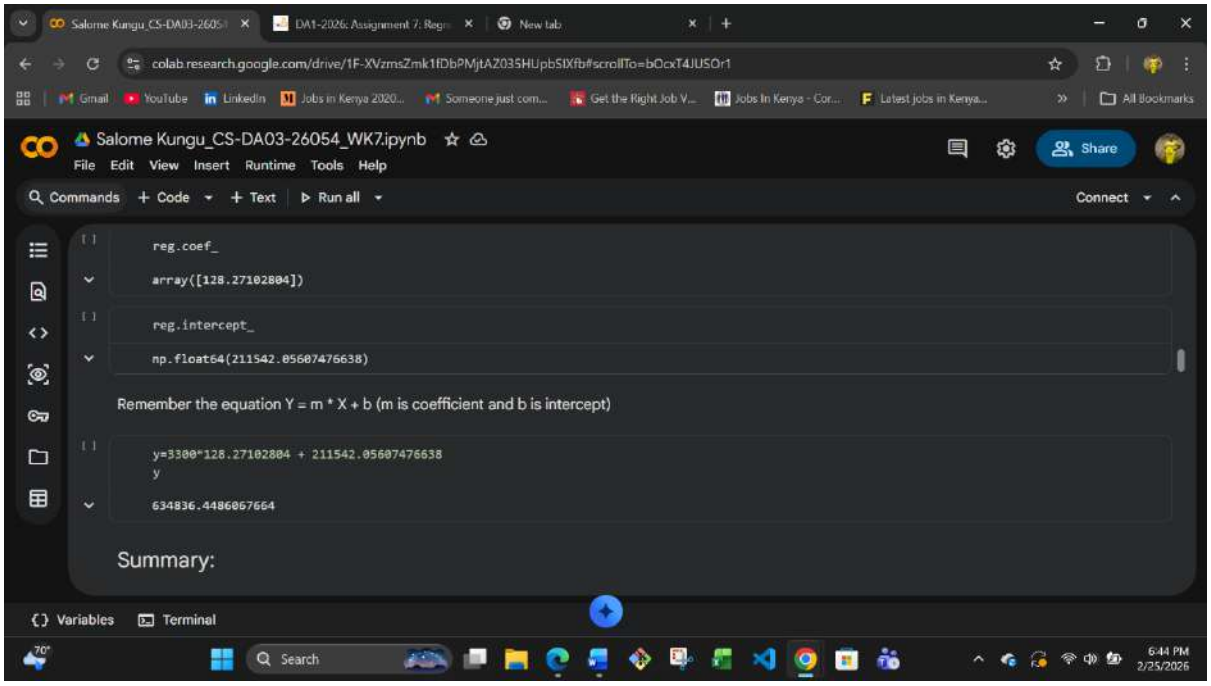




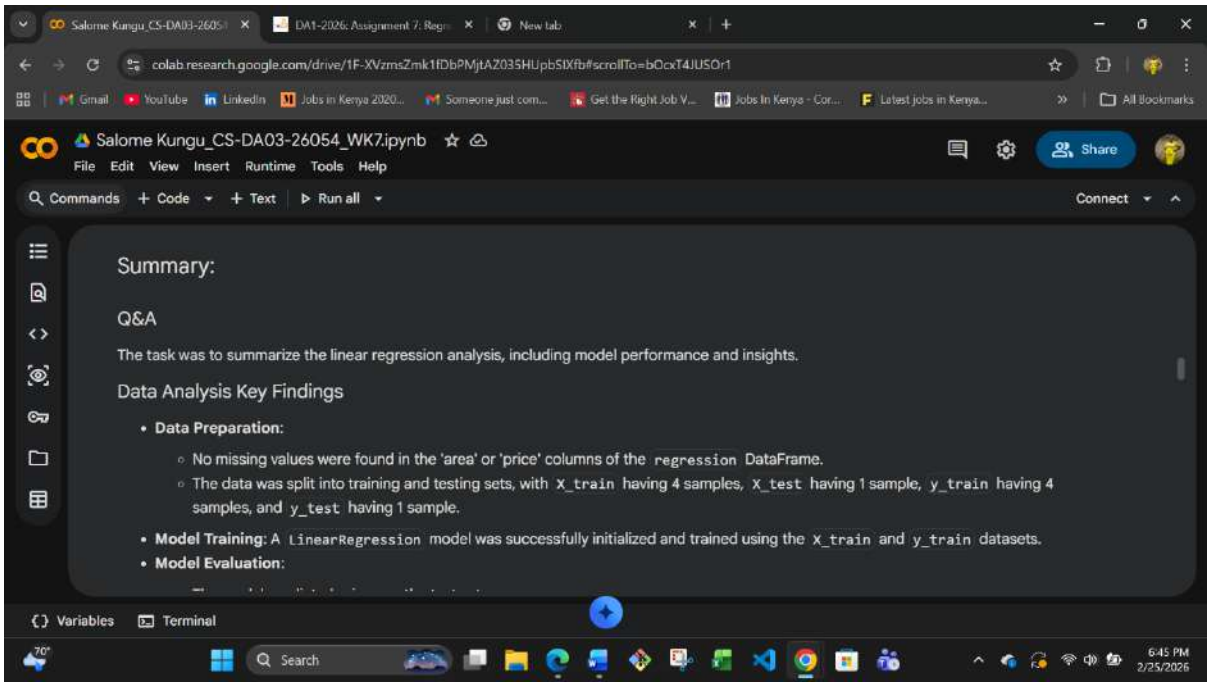


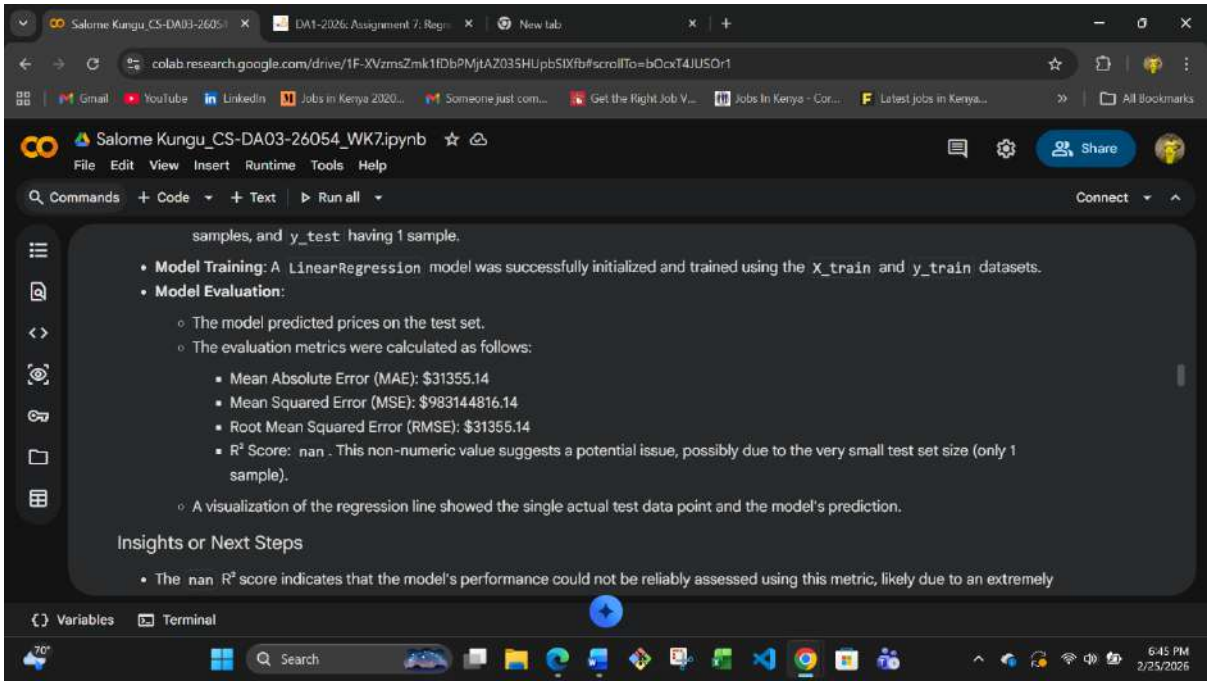
Example Of Liner Regression

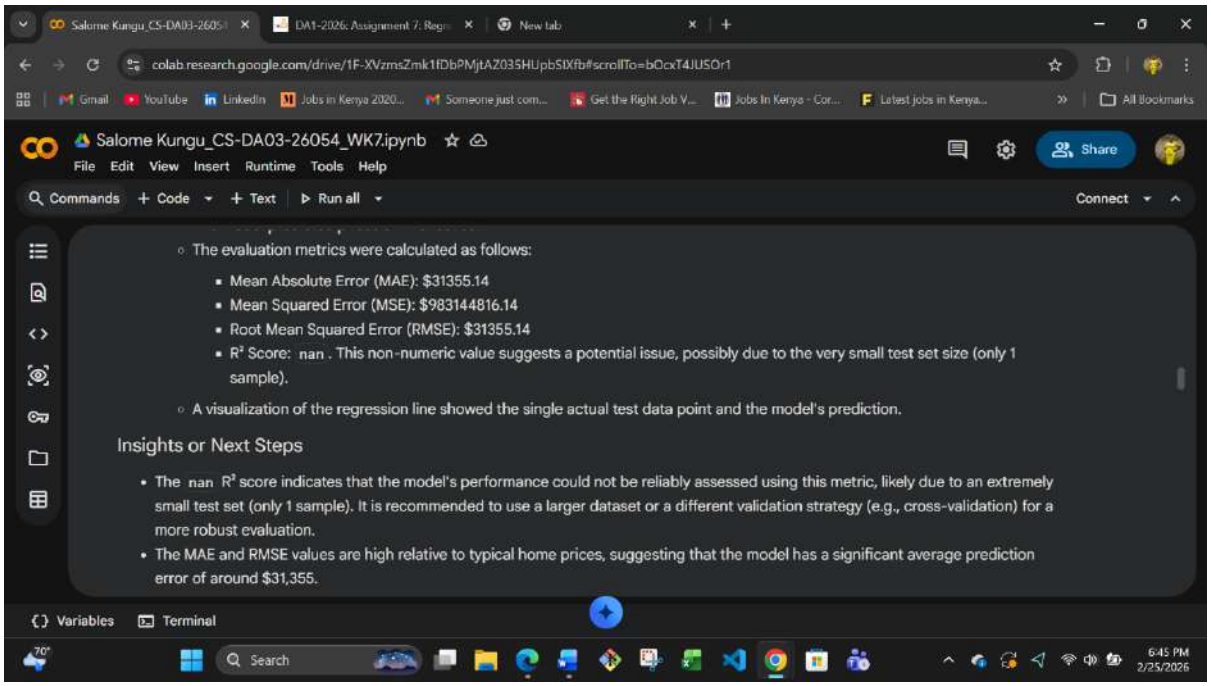




Summary Of Linear Regression







MULTIPLE LINEAR REGRESSION TASKS

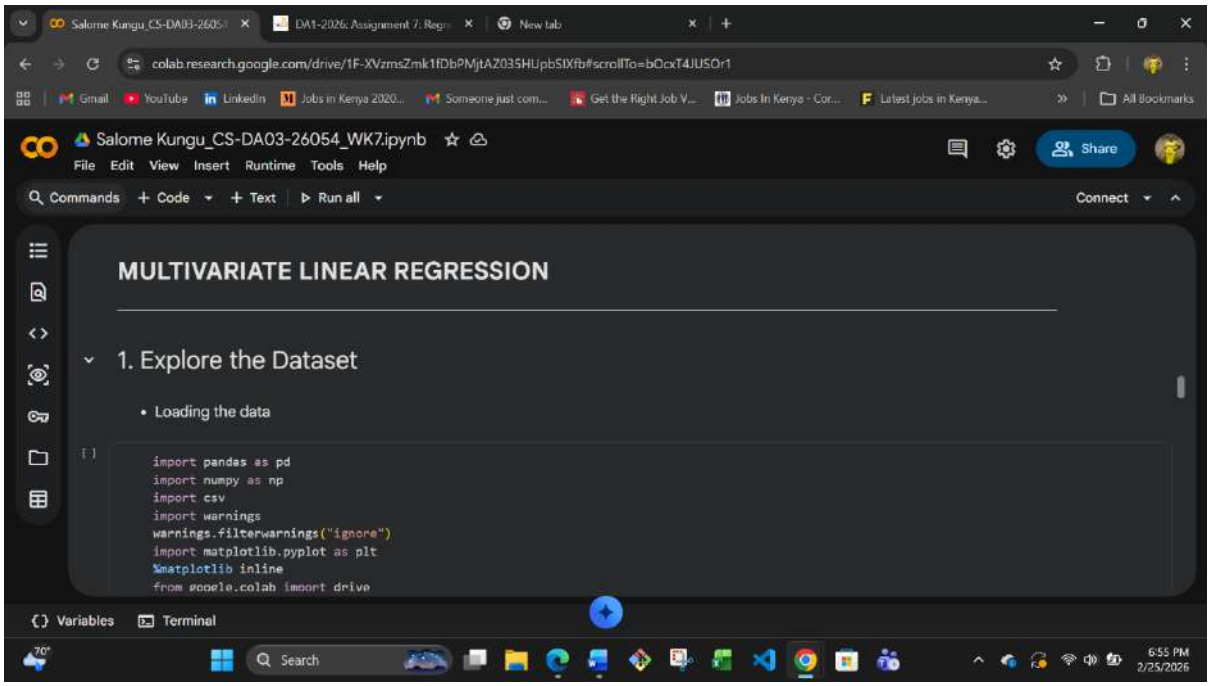
TASK 1: Data loading and exploration

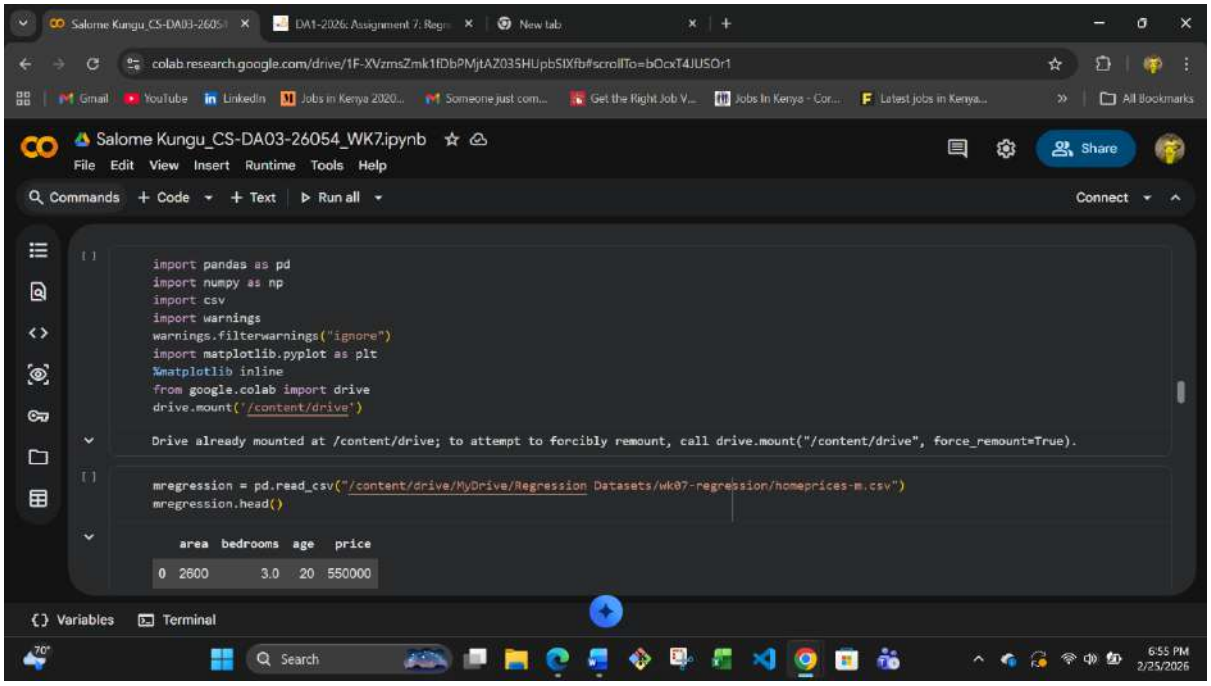
What:

- Select multiple independent variables
- Generate correlation matrix
- Check multicollinearity

Why:

- To evaluate feature relationships
- To detect redundant predictors
- To ensure model stability





Salome Kungu_CS-DA03-26054_WK7.ipynb

```
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
mregression = pd.read_csv("/content/drive/MyDrive/Regression Datasets/wk07-regression/homeprices-m.csv")
mregression.head()
```

	area	bedrooms	age	price
0	2600	3.0	20	550000
1	3000	4.0	15	565000
2	3200	NaN	18	610000
3	3600	3.0	30	595000
4	4000	5.0	8	760000

- Data Structure and Summary Statistics

Salome Kungu_CS-DA03-26054_WK7.ipynb

- Data Structure and Summary Statistics

```
print(" mregression DataFrame Information:")
regression.info()
```

```
 mregression DataFrame Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 4 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   area      6 non-null      int64
 1   bedrooms  6 non-null      float64
 2   age       6 non-null      int64
 3   price     6 non-null      int64
dtypes: float64(1), int64(3)
memory usage: 324.0 bytes
```

```
print("\nmregression DataFrame Descriptive Statistics:")
```

Salome Kungu_CS-DA03-26054_WK7.ipynb

```
print("\nmregression DataFrame Descriptive Statistics:")
regression.describe()
```

```
mregression DataFrame Descriptive Statistics:
```

	area	bedrooms	age	price
count	6.000000	6.00000	6.000000	6.000000
mean	3416.666667	4.20000	16.500000	648333.333333
std	587.934237	1.16619	8.288546	109117.673484
min	2600.000000	3.00000	8.000000	550000.000000
25%	3050.000000	3.25000	9.750000	572500.000000
50%	3400.000000	4.10000	16.500000	602500.000000
75%	3900.000000	4.80000	19.500000	722500.000000
max	4100.000000	6.00000	30.000000	810000.000000

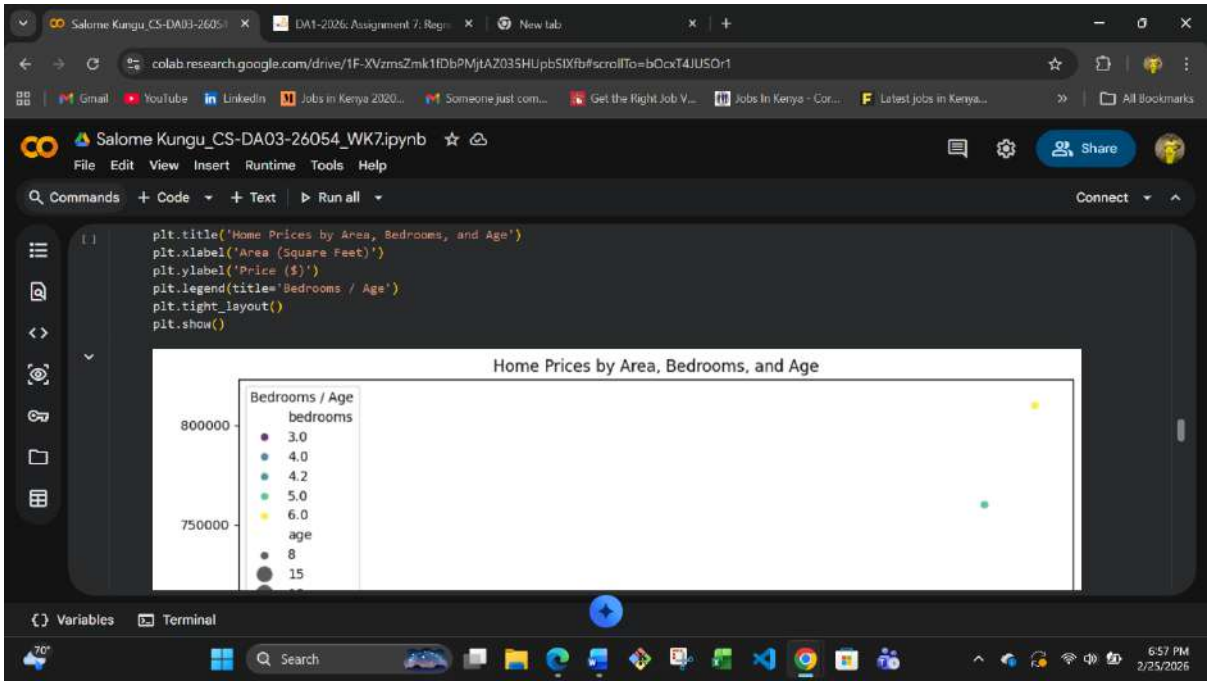
Salome Kungu_CS-DA03-26054_WK7.ipynb

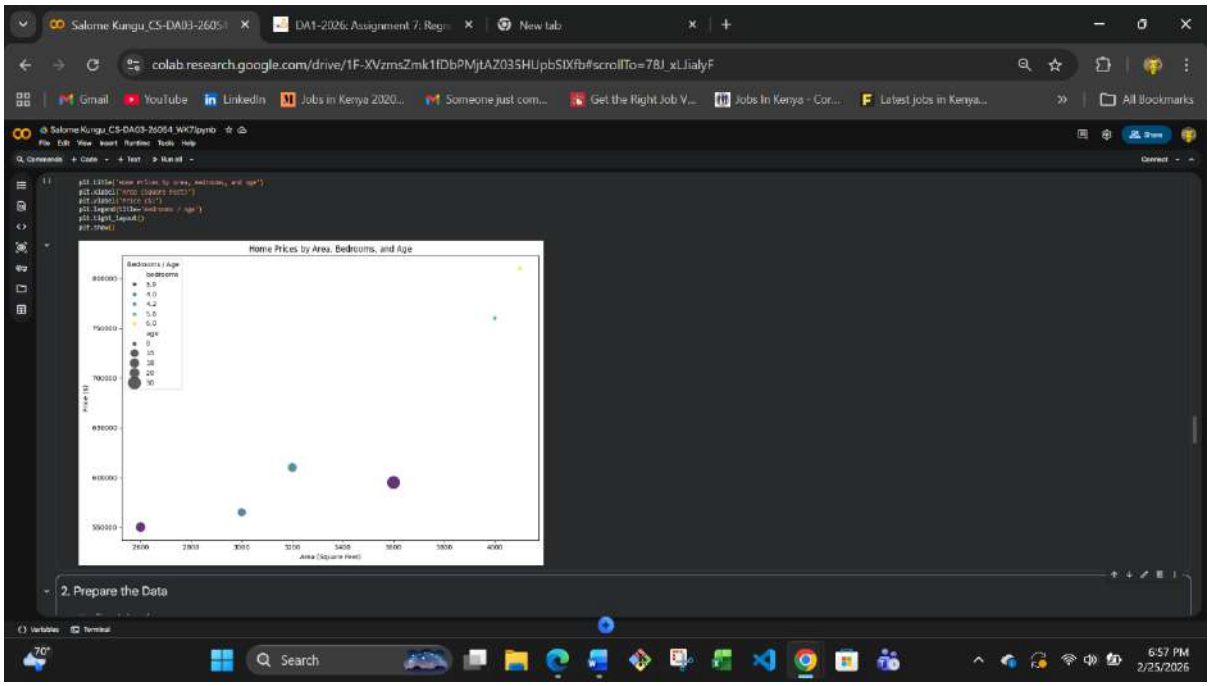
- Visualization using scatter plots

```
plt.figure(figsize=(10, 7))

sns.scatterplot(
    data=regression,
    x='area',
    y='price',
    hue='bedrooms',      # Color = bedrooms
    size='age',          # Size = age
    sizes=(40, 400),
    palette='viridis',
    alpha=0.8
)

plt.title('Home Prices by Area, Bedrooms, and Age')
plt.xlabel('Area (Square Feet)')
```





Task 2: Data Preparation for Multivariate Regression

What:

- Perform train-test split

Why:

- To include multiple predictors
- To capture combined feature effects
- To improve predictive capacity

Salome Kungu_CS-DA03-26054_WK7.ipynb

2. Prepare the Data

- Handling missing values

```
print('Missing values in the multi-variate regression DataFrame:')
print(mregression.isnull().sum())
```

```
Missing values in the multi-variate regression DataFrame:
area        0
bedrooms    1
age         0
price       0
dtype: int64
```

```
mregression['bedrooms'] = mregression['bedrooms'].fillna(
    mregression['bedrooms'].mean()
)
```

```
print('Missing values in the multi-variate regression DataFrame:')
print(mregression.isnull().sum())
```

```
Missing values in the multi-variate regression DataFrame:
area        0
bedrooms    0
age         0
price       0
dtype: int64
```

```
mregression['bedrooms'].fillna(mean_bedrooms, inplace=True)
print("Missing values in 'bedrooms' column filled with mean.")
mregression.head()
```

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```
mregression['bedrooms'].fillna(mean_bedrooms, inplace=True)
print("Missing values in 'bedrooms' column filled with mean.")
mregression.head()
```

```
Missing values in 'bedrooms' column filled with mean.
   area  bedrooms  age   price
0  2600       3.0   20  550000
1  3000       4.0   15  565000
2  3200       4.2   18  610000
3  3600       3.0   30  595000
4  4000       5.0    8  760000
```

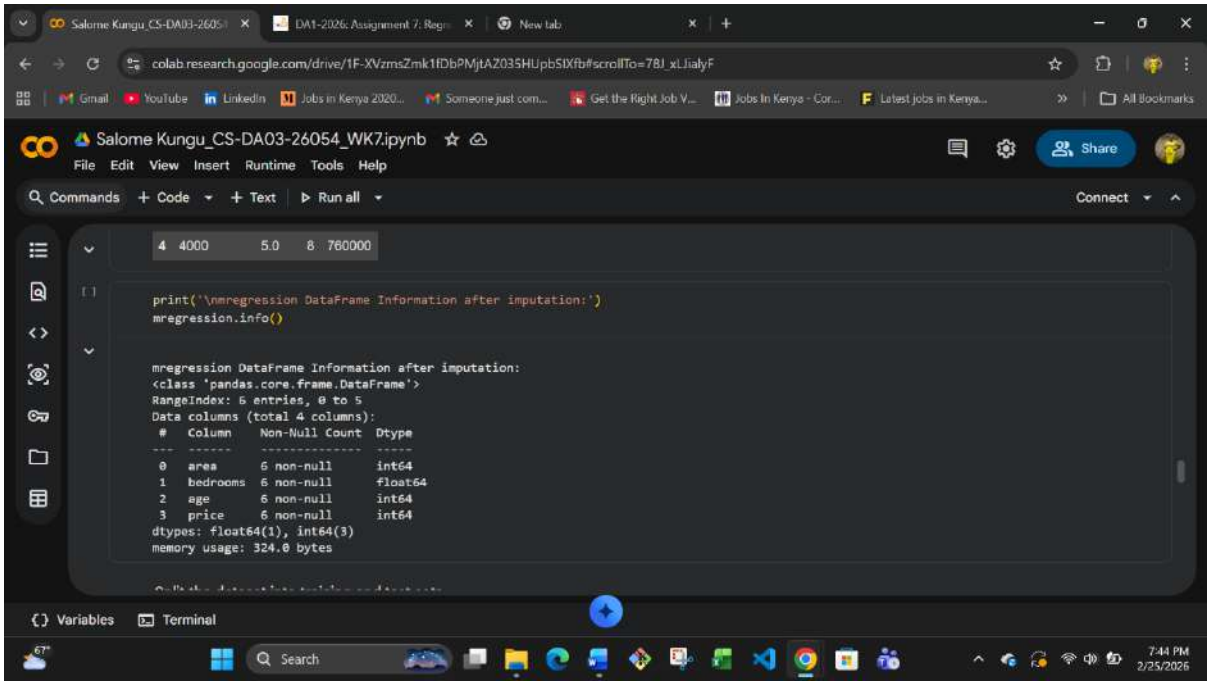
```
print('\nmregression DataFrame Information after imputation:')
mregression.info()
```

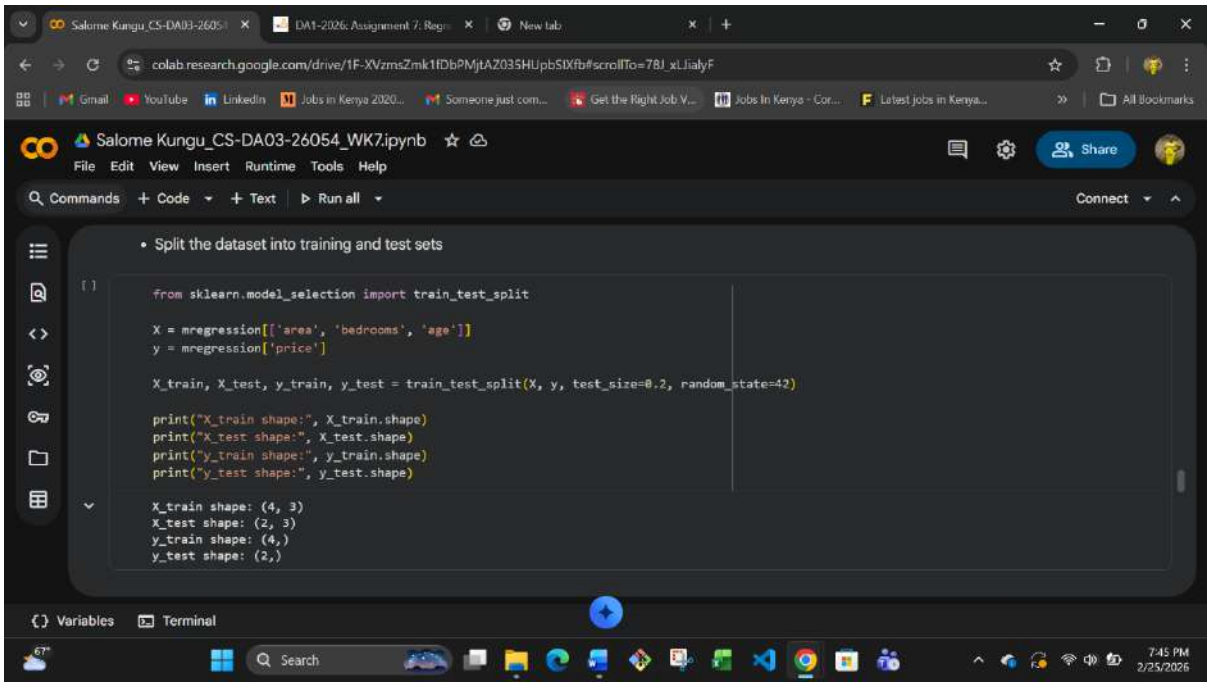
Salome Kungu_CS-DA03-26054_WK7.ipynb

```
4  4000       5.0    8  760000
```

```
print('\nmregression DataFrame Information after imputation:')
mregression.info()
```

```
mregression DataFrame Information after imputation:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 5
Data columns (total 4 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   area      6 non-null      int64
 1   bedrooms  6 non-null      float64
 2   age       6 non-null      int64
 3   price     6 non-null      int64
dtypes: float64(1), int64(3)
memory usage: 324.0 bytes
```





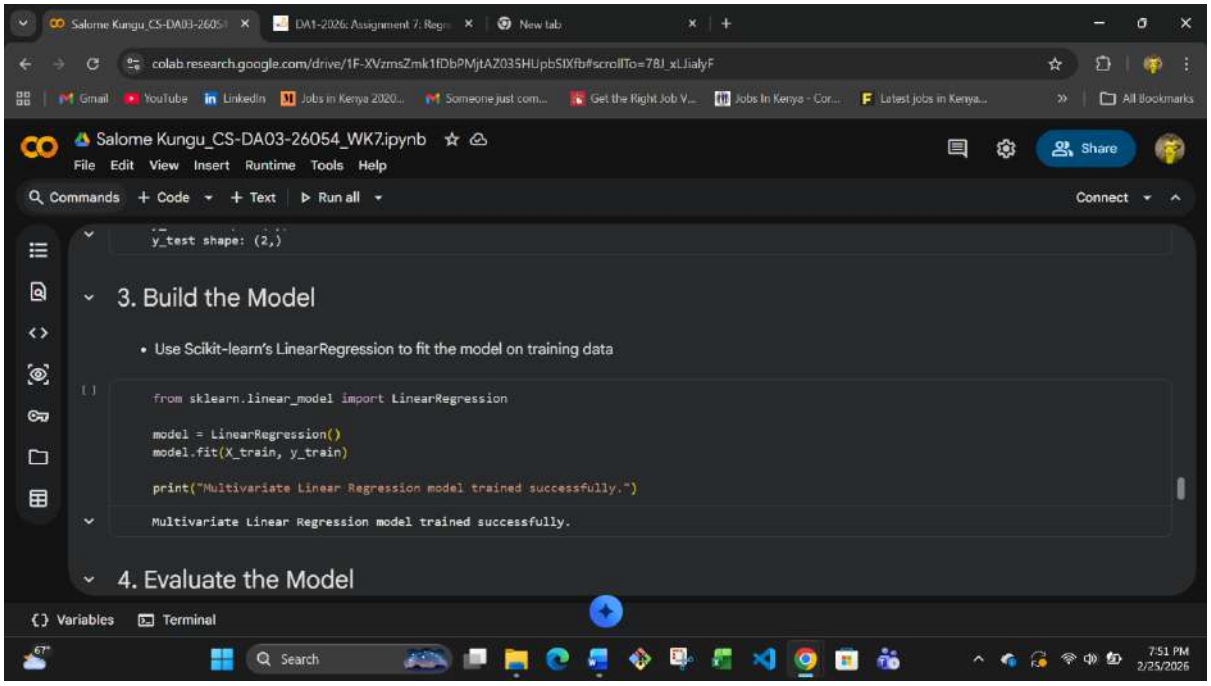
TASK 3: Train Multiple Linear Regression Model

What:

- Fit LinearRegression() with multiple variables
- Extract coefficients for each feature

Why:

- To quantify individual contribution of each predictor
- To model multivariate relationship
- To improve prediction accuracy



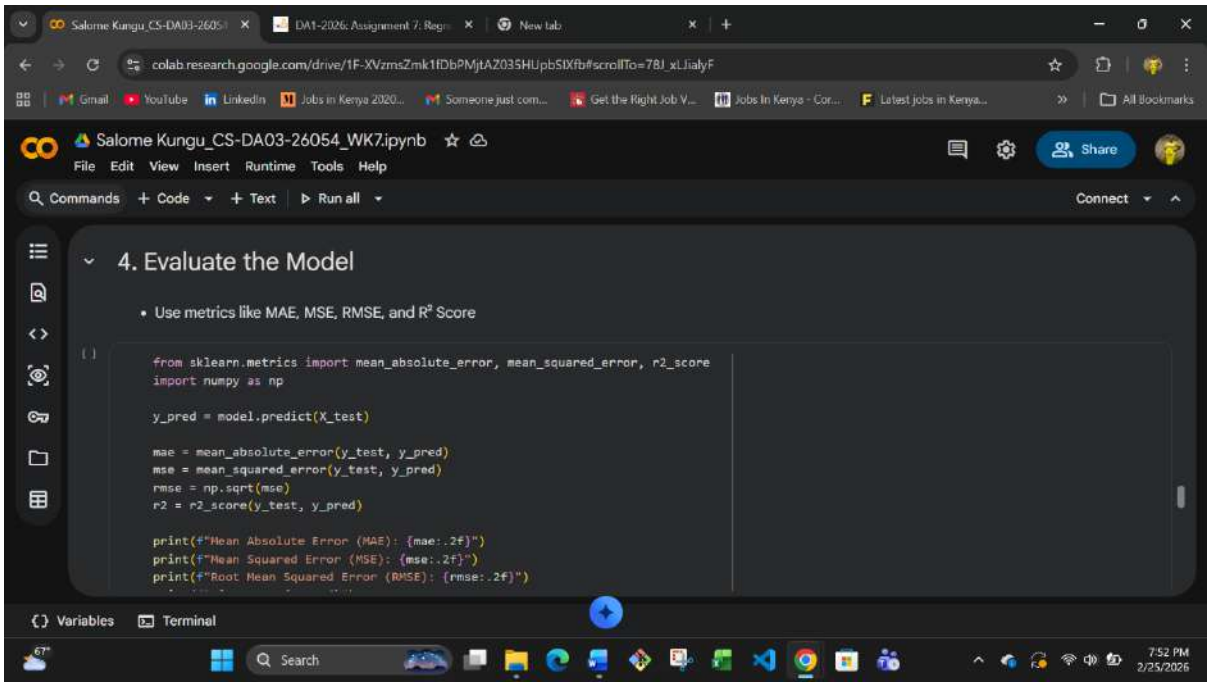
TASK 4: Evaluate Multiple Linear Regression

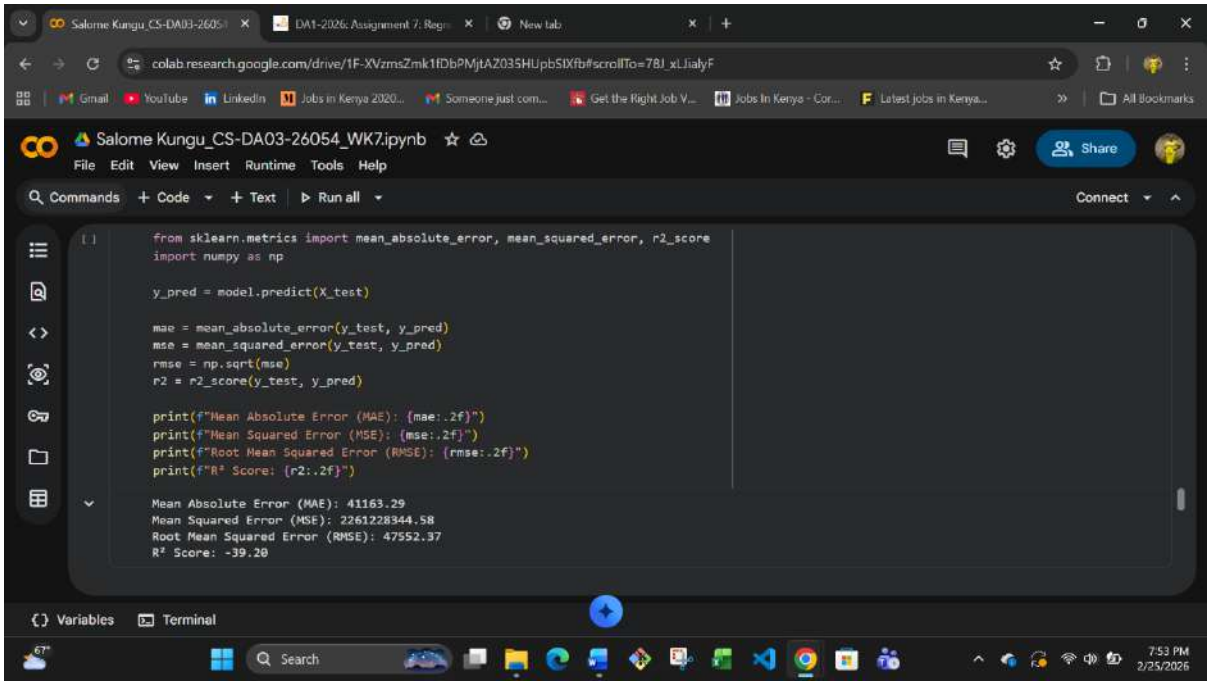
What:

- Calculate MAE, MSE, RMSE, R^2
- Compare with Simple Linear Regression results

Why:

- To determine performance improvement
- To assess explanatory enhancement
- To validate added feature usefulness





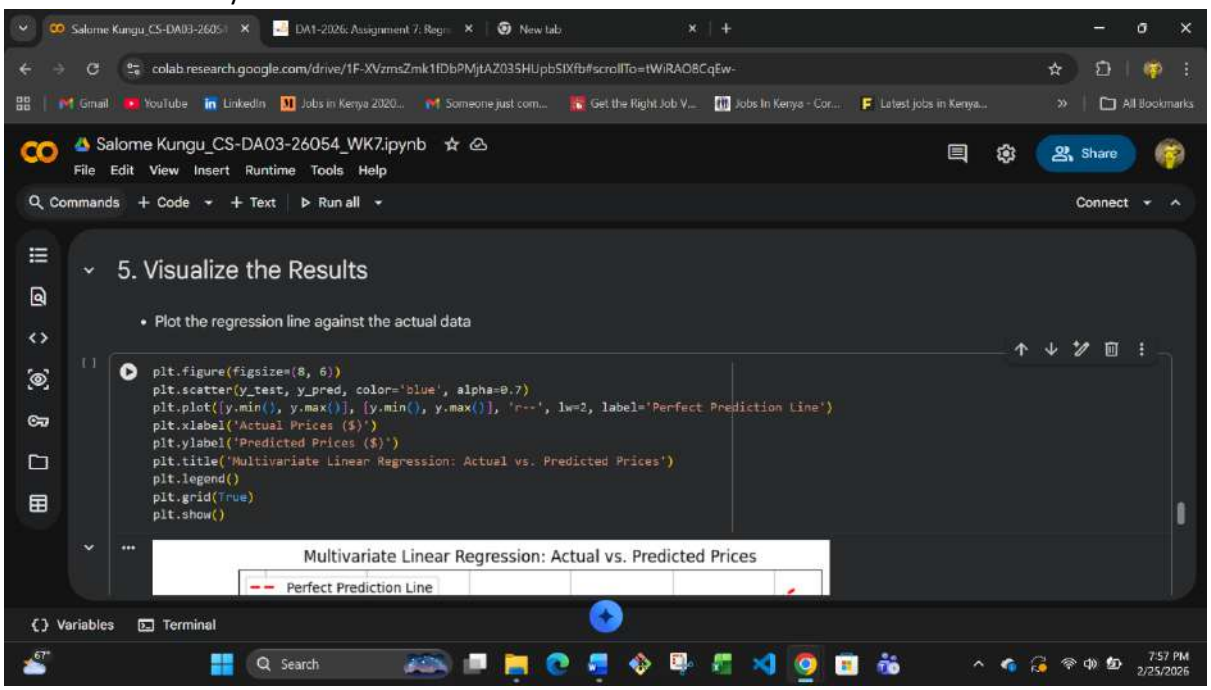
TASK 5: Visualize Multivariate Predictions

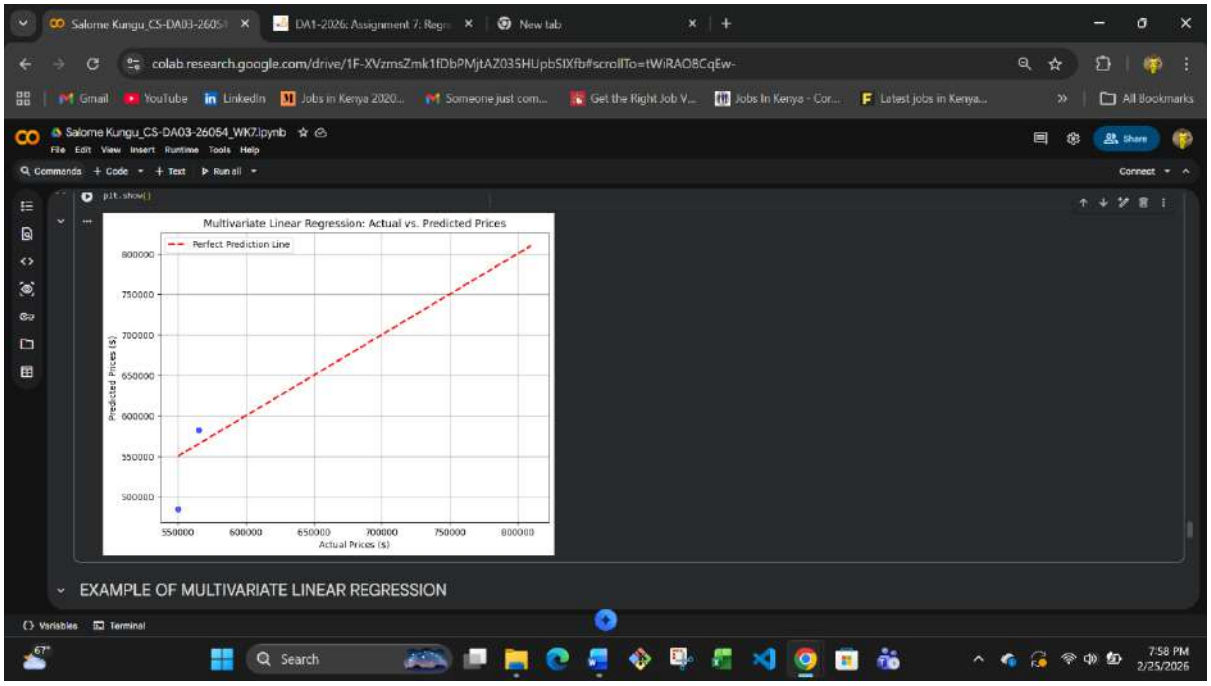
What:

- Plot Actual vs Predicted values
- Plot residual distribution

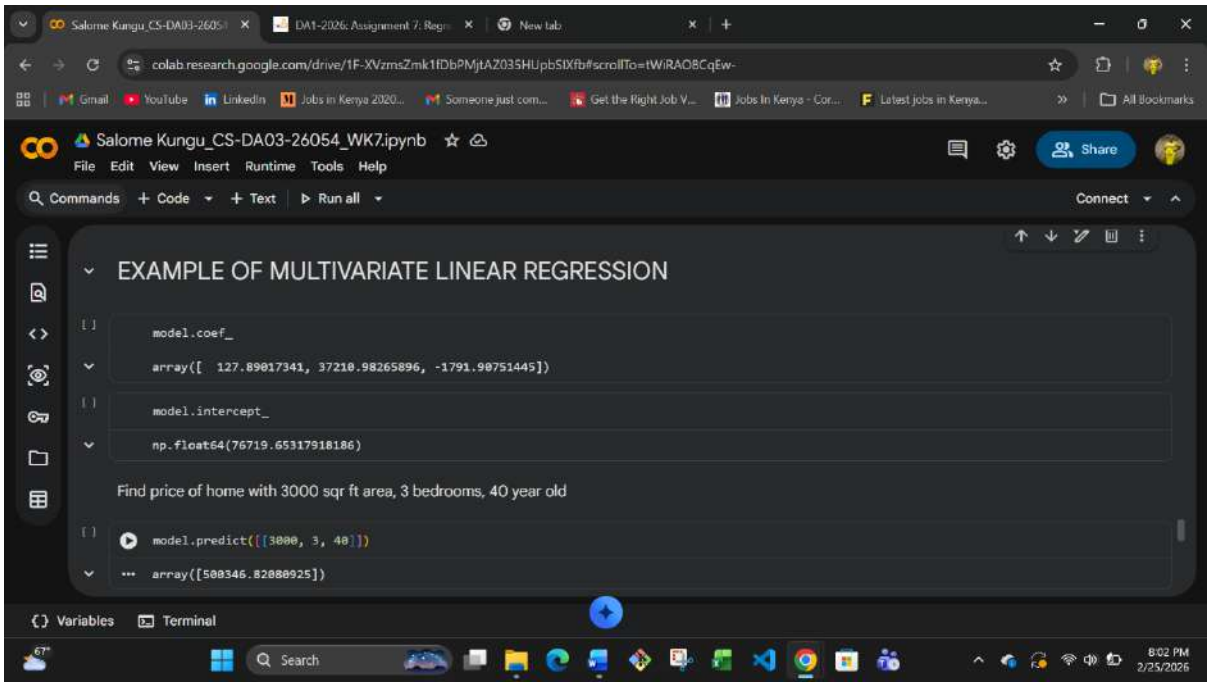
Why:

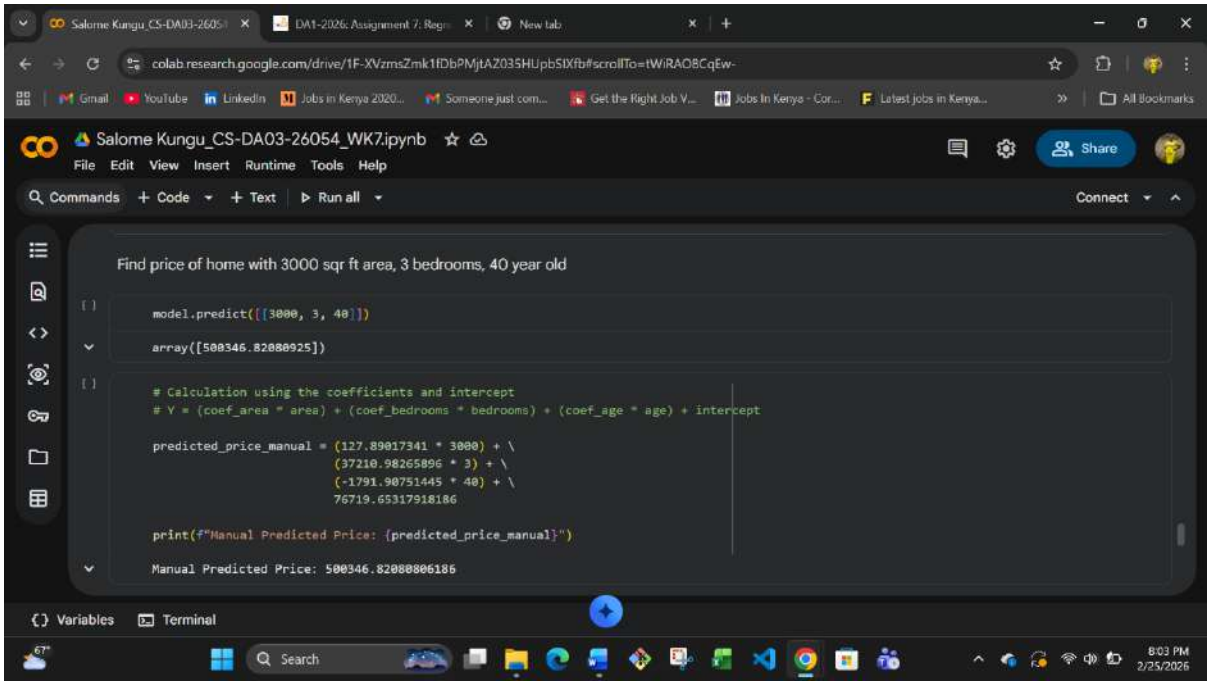
- To evaluate prediction dispersion
- To assess model accuracy visually
- To detect systematic bias

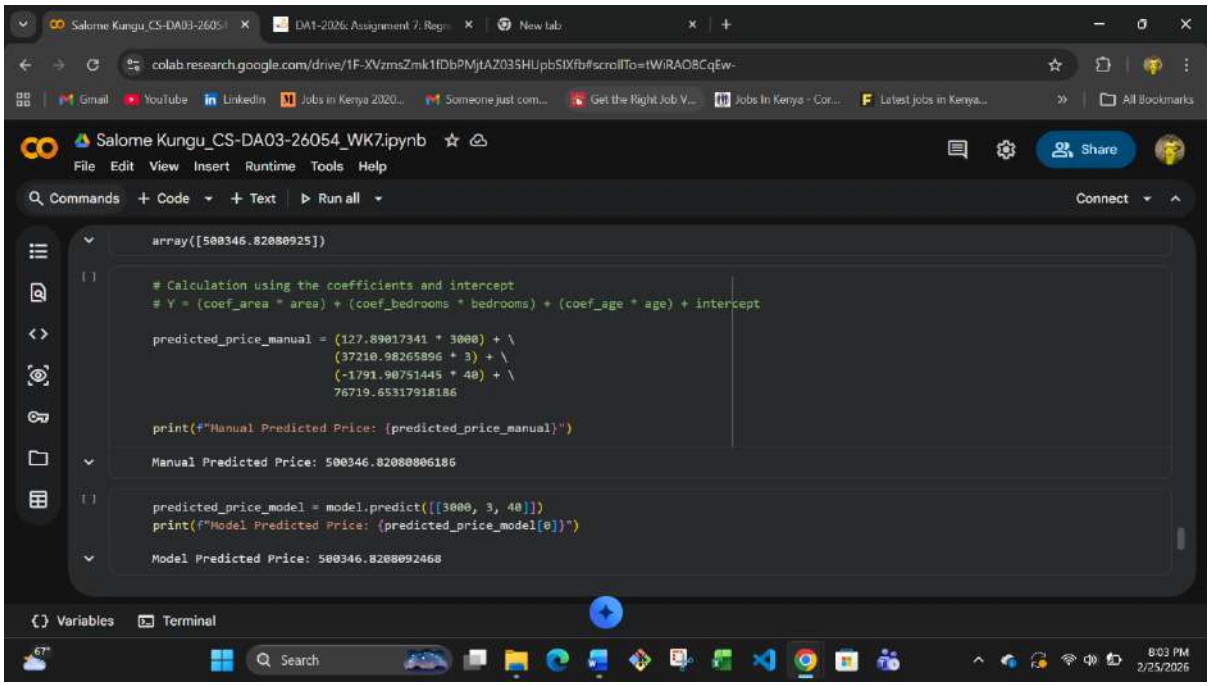




Example Of Multivariate Linear Regression







Summary Of Multivariate Linear Regression

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Summary:

Q&A

The task was to perform a multivariate linear regression analysis on the `homeprices-m.csv` dataset, including data exploration, confirmation of missing value handling, splitting the data into training and test sets with 'area', 'bedrooms', and 'age' as features and 'price' as the target, building and evaluating the linear regression model using metrics like MAE, MSE, RMSE, and R^2 Score, visualizing the actual versus predicted prices, and finally summarizing the findings and insights.

Data Analysis Key Findings

- **Data Loading & Initial Exploration**: The `homeprices-m.csv` dataset was loaded successfully, showing 'area', 'bedrooms', 'age', and 'price' columns. Initial `info()` and `describe()` calls revealed that 'bedrooms' had missing values.
- **Missing Value Handling**: One missing value in the 'bedrooms' column was identified and successfully imputed using the mean of the column (4.2). Subsequent checks confirmed no remaining missing values.

Salome Kungu_CS-DA03-26054_WK7.ipynb

Data Analysis Key Findings

- **Data Loading & Initial Exploration**: The `homeprices-m.csv` dataset was loaded successfully, showing 'area', 'bedrooms', 'age', and 'price' columns. Initial `info()` and `describe()` calls revealed that 'bedrooms' had missing values.
- **Missing Value Handling**: One missing value in the 'bedrooms' column was identified and successfully imputed using the mean of the column (4.2). Subsequent checks confirmed no remaining missing values.
- **Data Splitting**: The dataset was split into training and testing sets with 'area', 'bedrooms', and 'age' as features (`X`) and 'price' as the target (`y`). The training set (`X_train` , `y_train`) contained 4 samples, and the test set (`X_test` , `y_test`) contained 2 samples.

Model Training & Evaluation

- **Model Building**: A `LinearRegression` model from Scikit-learn was successfully initialized and trained using the `X_train` and `y_train` datasets.
- **Model Evaluation**: The model's performance on the test set was evaluated using several metrics:

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- **Model Evaluation**: The model's performance on the test set was evaluated using several metrics:
  - Mean Absolute Error (MAE): $41163.29 * MeanSquaredError(MSE)$:2261228344.58
  - Root Mean Squared Error (RMSE): $47552.37
  - R^2 Score: -39.20

Insights or Next Steps

- **R^2 Score Interpretation**: The R^2 Score of -39.20 is highly problematic. An R^2 score less than 0 indicates that the model performs worse than simply predicting the mean of the target variable. This extreme negative value suggests that the model is a very poor fit for the data, especially given the very small test set size (only 2 samples). This could lead to a highly unstable and unrepresentative R^2 score.
- **Error Magnitudes**: The MAE, MSE, and RMSE values are quite high, indicating significant prediction errors (e.g., an average absolute

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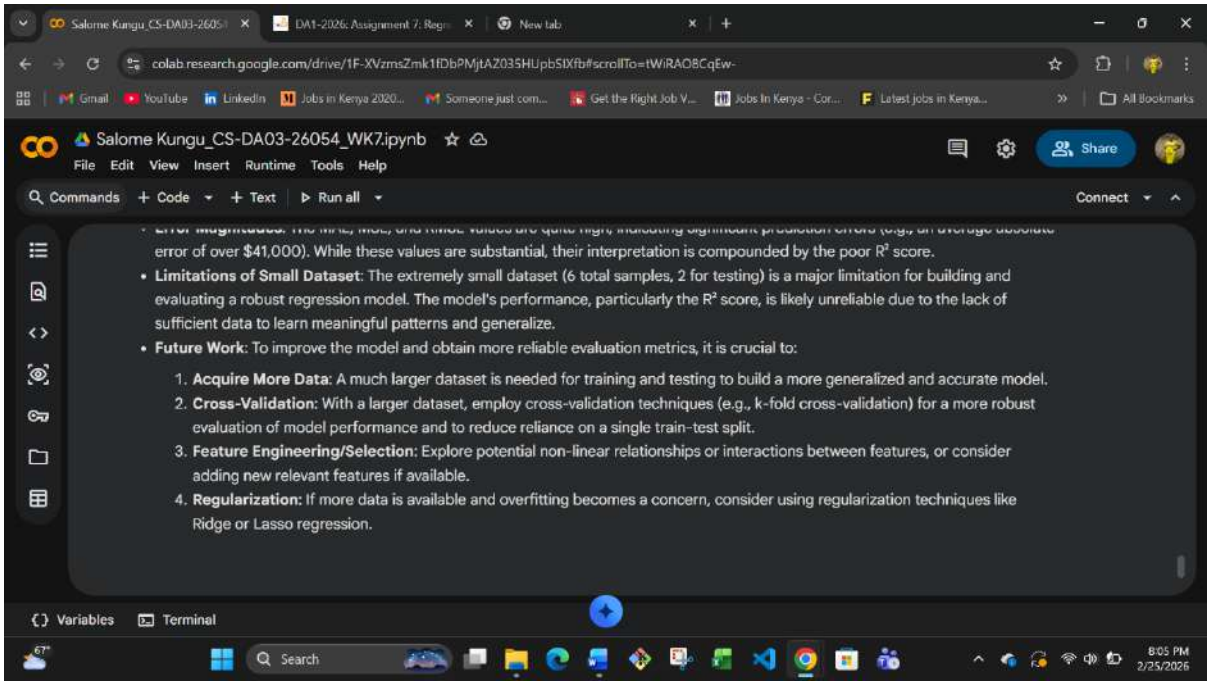
- **Error Magnitudes**: The MAE, MSE, and RMSE values are quite high, indicating significant prediction errors (e.g., an average absolute error of over $41,000). While these values are substantial, their interpretation is compounded by the poor R^2 score.
- **Limitations of Small Dataset**: The extremely small dataset (6 total samples, 2 for testing) is a major limitation for building and evaluating a robust regression model. The model's performance, particularly the R^2 score, is likely unreliable due to the lack of sufficient data to learn meaningful patterns and generalize.
- **Future Work**: To improve the model and obtain more reliable evaluation metrics, it is crucial to:
  1. **Acquire More Data**: A much larger dataset is needed for training and testing to build a more generalized and accurate model.



Conclusion

This assignment provided comprehensive hands-on experience in implementing both **Simple Linear Regression and Multiple Linear Regression** using Python.

Through this project, I strengthened my understanding of:

- Supervised learning principles
- Feature selection and importance
- Model evaluation techniques
- Comparing baseline vs enhanced models
- Translating mathematical regression concepts into practical implementation

The comparison between single-feature and multi-feature regression demonstrated how incorporating additional relevant variables significantly improves predictive performance.

The completed notebook serves as a strong addition to my data analytics and machine learning portfolio, showcasing my ability to build, evaluate, and compare predictive models professionally.